

MODULE - 05

CURVE FITTING, CORRELATION AND REGRESSIONS

* CURVE FITTING

This is a method of finding a specific relation connecting the dependent and independent variables for a given data so as to satisfy the data as accurately as possible.

* Curve-fitting by the method of least squares.

Curve fitting is a method of finding a suitable relation or law in the form of $y = f(x)$ for a set of observed values (x_i, y_i) , $i = 1, 2, \dots, n$.

Suppose, $y = f(x)$ is an appropriate value of the relation that fit into the given data comparing (x_i, y_i) for $i = 1, 2, \dots, n$. then

y_i are called observed values,

$y_i = f(x_i)$ are called expected values.

Their difference $R_i = y_i - y_i$ is called residual or estimated errors.

∴ The method of least square provides a relationship $y = f(x)$ such that the sum of square of residuals is the least.

(2)

* To fit a straight line $y = ax + b$.

Consider the curve of the form $y = ax + b$.
where a & b are parameters.

Consider a set of n given values (x, y) for fitting the straight line $y = ax + b$.

The residuals $R = y - (ax + b)$ is the difference between the observed & estimated values of y by the method of least square, we find the parameters a & b such that the sum of square of residuals is minimum.

$$\text{Let } S = \sum_1^n R^2$$

$$S = \sum_1^n [y - (ax + b)]^2$$

Treating S as a function of two parameters a & b the necessary conditions for S to be minimum are

$$\frac{\partial S}{\partial a} = 0 \quad \& \quad \frac{\partial S}{\partial b} = 0. \quad \text{i.e. partially differentiating w.r.t } a \& b$$

$$\text{i.e. } \frac{\partial S}{\partial a} = 2 \sum_1^n [y - (ax + b)](-x) = 0$$

$$\frac{\partial S}{\partial b} = 2 \sum_1^n [y - (ax + b)](-1) = 0$$

Dividing both by 2 we have

$$\frac{\partial S}{\partial a} = \sum_1^n [xy + ax^2 + bx] = 0$$

$$\frac{\partial S}{\partial b} = \sum_1^n [-y + ax + b]$$

$$\left[\because \sum_1^n b = b + b + \dots + n \text{ times} \right. \\ \left. = nb \right]$$

∴ the above equations becomes.

$$\begin{cases} \Sigma y = a \Sigma x + bn \\ \Sigma xy = a \Sigma x^2 + b \Sigma x \end{cases}$$

These equations are called the normal equations.

and on solving we get the values of a & b .

So that $y = ax + b$ is the equation of the best fitting straight line in the least square sense.

* To fit a parabola of second degree $y = ax^2 + bx + c$.

Consider a set of n pairs of given values (x, y) for fitting the curve $y = ax^2 + bx + c$.

The residual $R = y - (ax^2 + bx + c)$ is the difference between the observed & estimated value of y .

$$\text{Let } S = \sum_1^n [y - (ax^2 + bx + c)]^2$$

Treating S as a fⁿ of three parameters a, b, c the necessary condition for S to be minimum are $\frac{\partial S}{\partial a} = 0, \frac{\partial S}{\partial b} = 0, \frac{\partial S}{\partial c} = 0$.

$$\text{i.e. } \frac{\partial S}{\partial a} = 2 \sum_1^n [y - (ax^2 + bx + c)] (-x^2) = 0.$$

$$\frac{\partial S}{\partial b} = 2 \sum_1^n [y - (ax^2 + bx + c)] (-x) = 0$$

$$\frac{\partial S}{\partial c} = 2 \sum_1^n [y - (ax^2 + bx + c)] (-1) = 0$$

÷ above equations by 2 we have,

$$-\sum_1^n x^2 y + \sum_1^n a x^4 + \sum_1^n b x^3 + \sum_1^n c x^2 = 0$$

$$-\sum_1^n x y + \sum_1^n a x^3 + \sum_1^n b x^2 + \sum_1^n c x = 0$$

$$-\sum_1^n y + \sum_1^n a x^2 + \sum_1^n b x + \sum_1^n c = 0.$$

But $\sum_1^n c = c + c + c + \dots + n = n c$

∴ the above equations becomes

$$\therefore a \sum y = a \sum x^2 + b \sum x + n c$$

$$\sum x y = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

These equations are called normal equations. & on solving we get the values of a, b, c so that $y = ax^2 + bx + c$ is the best fitting parabola of second degree in the least square sense.

* To fit a curve of the form $y = ax^b$

consider, $y = ax^b$

Taking log on b.s.

$$\log_e y = \log_e a + b \log_e x \quad (1)$$

$$Y = A + BX \quad \text{--- (1)}$$

where $Y = \log_e y$, $A = \log_e a$, $B = b$, $X = \log_e x$

1. The normal equations associated with (1) are as follows.

$$\begin{aligned} \sum Y &= nA + B \sum X & \text{--- (2)} \\ \sum XY &= A \sum X + B \sum X^2 & \text{--- (3)} \end{aligned}$$

Solving (2) & (3) we obtain A, B.

But, $\log_e a = A \Rightarrow a = e^A$

Sub the values of a, b in $y = ax^b$
we get the curve of best fit in required form.

* note:- we can similarly fit the curves
 $y = ae^{bx}$, $y = ab^x$ etc.

* NOTE:-

The normal equations for fitting a straight line or parabola can be written instantly from the desired equation of the curve as follows.

- we 1st apply summation $\sum$ to the desired equation keeping the constants a, b, c outside the summation where the summation of pure constant terms like $\sum a$, $\sum b$, $\sum c$ are to be written as na , nb , nc -- respectively.
- we then multiply the given eqn by independent variable x & apply summation again. This will suffice for fitting a straight line. However in the case of parabola we must also multiply by x^2 & apply summation.

* Working Procedure for problems

Step 1: we first write the normal equations appropriate to the curve of best fit.

Step 2: we prepare the relevant table & find the value of all the summations present in the normal equations

Step 3: we substitute these values to arrive at a system of equations in the unknown parameters.

Step 4: we find the parameters by solving the system of equations & substitute in the equation $y = f(x)$ which being the curve of best fit.

* NOTE:

- ① The solution of the system of equations in 2 or 3 unknown can be obtained directly from the latest commonly used scientific calculators.
- ② If the magnitudes involved in the data are big, we may modify the data choosing a convenient origin somewhere in the middle & finally revert back.

* PROBLEMS

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① Fit a straight line $y = ax + b$ for the following data.

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

→ The normal equations for fitting the straight line $y = ax + b$ are

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x \quad (n=8)$$

The relevant table is as follows.

x	y	xy	x^2
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
$\sum x = 56$	$\sum y = 40$	$\sum xy = 364$	$\sum x^2 = 524$

The normal equations becomes,

$$56a + 8b = 40$$

$$524a + 56b = 364$$

On solving the equations, we have,

$$a = 0.636363636$$

$$\boxed{a \approx 0.64}$$

$$b = 0.545454545$$

$$\boxed{b = 0.55}$$

Thus by substituting these values in

$y = ax + b$, we obtain

$$\boxed{y = 0.64x + 0.55} //$$

② Find the equation of the best fitting straight line for the following data and hence estimate the value of the dependent variable corresponding to the value of 30 of the independent variable.

x	5	10	15	20	25
y	16	19	23	26	30

Let $y = ax + b$ be the equation of best fitting straight line.

The normal equations are as follows.

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x \quad (n=5)$$

The relevant table is as follows.

x	y	xy	x^2
5	16	80	25
10	19	190	100
15	23	345	225
20	26	520	400
25	30	750	625
$\Sigma x = 75$	$\Sigma y = 114$	$\Sigma xy = 1885$	$\Sigma x^2 = 1375$

The normal equations becomes,

$$\Sigma y = a \Sigma x + nb$$
$$114 = 75a + 5b \quad \text{--- (1)}$$

$$\Sigma xy = a \Sigma x^2 + b \Sigma x$$
$$1885 = 1375a + 75b \quad \text{--- (2)}$$

Solving, we have,

$$a = 0.7, \quad b = 12.3$$

Thus by fitting straight line in the form.

$$y = 0.7x + 12.3$$

Further, when $x = 30$

$$\text{we obtain, } y = 0.7(30) + 12.3$$

$$y = 21 + 12.3$$

$$\boxed{y = 33.3} //$$

(10)

③ Fit a straight line to the following data.

Year	1961	1971	1981	1991	2001
production (in tons)	8	10	12	10	16

Also find the expected production in the year 2006.

→ Let year & production respectively be represented by the variables x & y .

We shall fit the straight line in the form $y = a + bx$

Since the values of x are large in magnitude, we may prefer to modify the same by choosing convenient origin somewhere in middle.

Let $X = x - 1981$ & the line of fit will be $y = a + bX$

[note: If the data was upto 1991 (say) we can take $X = x - 1975$]

The normal equations associated with $y = a + bX$ are as follows.

$$\sum y = na + b \sum X$$

$$\sum Xy = a \sum X + b \sum X^2 \quad (n=5)$$

The relevant table is as follows.

$X = (x - 1981)$	y	Xy	X^2
-20	8	-160	400
-10	10	-100	100
0	12	0	0
10	10	100	100
20	16	320	400
$\Sigma X = 0$	$\Sigma y = 56$	$\Sigma Xy = 160$	$\Sigma X^2 = 1000$

The normal equations become,

$$\Sigma y = na + b \Sigma X \quad ; \quad \Sigma Xy = a \Sigma X + b \Sigma X^2$$

$$56 = 5a + b(0) \quad 160 = a(0) + b(1000)$$

$$56 = 5a$$

$$160 = 1000b$$

Solving,

$$\therefore a = 11.2 \quad \& \quad b = 0.16$$

Hence $y = a + bX$ with $X = x - 1981$ becomes

$$y = 11.2 + 0.16(x - 1981)$$

Thus, $y = -305.76 + 0.16x$ is the required line to fit.

Also when $x = 2006$,

$$y = -305.76 + 0.16(2006)$$

$$y = 15.2$$

Expected production in the year 2006 is 15.2 tons

④ Fit a second degree parabola $y = ax^2 + bx + c$ in the least square sense for the following data & hence estimate y at $x = 6$.

x	1	2	3	4	5
y	10	12	13	16	19

→ The normal equations associated with $y = ax^2 + bx + c$ are as follows.

$$\sum y = a \sum x^2 + b \sum x + nc$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2 \quad (n=5)$$

The relevant table is as follows.

x	y	xy	x^2	x^3	x^4
1	10	10	1	1	1
2	12	24	4	8	16
3	13	39	9	27	81
4	16	64	16	64	256
5	19	95	25	125	625
$\sum x = 15$	$\sum y = 70$	$\sum xy = 232$	$\sum x^2 = 55$	$\sum x^3 = 225$	$\sum x^4 = 979$

The normal equations becomes.

$$70 = 55a + 15b + 5c$$

$$232 = 225a + 85b + 15c$$

$$979a = 979a + 225b + 55c$$

On solving we have,

$$a = 0.2857$$

$$b = 0.4857$$

$$c = 9.4$$

$$a \approx 0.29$$

$$b \approx 0.49$$

Thus the required second degree parabola is

$$y = 0.29x^2 + 0.49x + 9.4$$

Also at $x = 6$

$$y = 0.29(36) + 0.49(6) + 9.4$$

$$y = 22.78$$

⑤ Fit a parabola $y = a + bx + cx^2$ for the data:

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

→ The normal equations associated with $y = a + bx + cx^2$ are as follows.

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \quad (n=5)$$

The relevant table as follows.

x	y	xy	x^2y	x^2	x^3	x^4
0	1	0	0	0	0	0
1	1.8	1.8	1.8	1	1	1
2	1.3	2.6	5.2	4	8	16
3	2.5	7.5	22.5	9	27	81
4	6.3	25.2	100.8	16	64	256
$\Sigma x = 10$	$\Sigma y = 12.9$	$\Sigma xy = 37.1$	$\Sigma x^2y = 130.3$	$\Sigma x^2 = 30$	$\Sigma x^3 = 100$	$\Sigma x^4 = 354$

The normal equations becomes,

$$12.9 = 5a + 10b + 30c$$

$$37.1 = 10a + 30b + 100c$$

$$130.3 = 30a + 100b + 354c$$

On solving we have,

$$a = 1.42, b = -1.07, c = 0.55$$

Then the parabola of fit is,

$$\boxed{y = 1.42 - (1.07)x + 0.55x^2} //$$

- ⑥ Find the best values of a, b, c if the equation $y = a + bx + cx^2$ is to fit most closely to following observations.

x	-2	-1	0	1	2
y	-3.150	-1.390	0.620	2.880	5.378

→ The normal equations associated with

$y = a + bx + cx^2$ are as follows.

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \quad (n=5)$$

The relevant table is as follows.

x	y	xy	$x^2 y$	x^2	x^3	x^4
-2	-3.15	-6.30	-12.60	4	-8	16
-1	-1.39	-1.39	-1.39	1	-1	1
0	0.62	0	0	0	0	0
1	2.88	2.88	2.88	1	1	1
2	5.378	10.756	21.512	4	8	16
$\sum x = 0$	$\sum y = 4.338$	$\sum xy = 21.326$	$\sum x^2 y = 10.402$	$\sum x^2 = 10$	$\sum x^3 = 0$	$\sum x^4 = 34$

The normal equations becomes,

$$4.338 = 5a + 10c \quad \rightarrow (1)$$

$$21.326 = 10b \quad \rightarrow (2)$$

$$10.402 = 10a + 34c \quad \rightarrow (3)$$

On solving, eqn (2) we get

$$b = 2.1326$$

eqn (1) & (3) we get, $a = 0.6210$, $c = 0.1233$

Thus the best values a, b, c for fitting parabola $y = a + bx + cx^2$ are

$$a = 0.621, \quad b = 2.1326, \quad c = 0.1233$$

$$y = 0.621 + 2.1326x + 0.1233x^2$$

④ Fit a least square geometric curve $y = ax^b$ for the following data.

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

→ Consider, $y = ax^b$

$$\therefore \log_e y = \log_e a + b \log_e x \quad \&$$

$$\text{let } Y = \log_e y, \quad A = \log_e a, \quad X = \log_e x$$

The normal equations associated with $Y = A + bX$ are as follows.

$$\sum Y = nA + b \sum X$$

$$\sum XY = A \sum X + b \sum X^2 \quad (n=5)$$

The relevant table is as follows.

x	y	$X = \log_e x$	$Y = \log_e y$	XY	X^2
1	0.5	0	-0.6931	0	0
2	2	0.6931	0.6931	0.4804	0.4804
3	4.5	1.0986	1.5041	1.6524	1.2069
4	8	1.3863	2.0794	2.8827	1.9218
5	12.5	1.6094	2.5257	4.0649	2.5902
		$\sum X = 4.7874$	$\sum Y = 6.1092$	$\sum XY = 9.0804$	$\sum X^2 = 6.1993$

The normal equations become,

$$6.1092 = 5A + 4.7874b$$

$$9.0804 = 4.787A + 6.1993b$$

On solving we have, $a = -0.69315$, $b = 2$

$$\log_e a = a \Rightarrow a = e^a = e^{-0.69315} = 0.5$$

Then, $y = ax^b$

$y = 0.5x^2$ is the required curve of fit.

⑧ An experiment on lifetime t of cutting tool at different cutting speeds v (units) are given below.

Speed (v)	350	400	500	600
Life (t)	61	26	7	2.6

Fit a relation of the form $v = at^b$

→ Consider, $v = at^b$

$$\log_e v = \log_e a + b \log_e t$$

Let $V = \log_e v$, $A = \log_e a$, $T = \log_e t$ and hence we have the equation $V = A + bT$

The normal equations are as follows

$$\sum V = nA + b \sum T$$

$$\sum VT = A \sum T + b \sum T^2 \quad (n = 4)$$

The relevant table is as follows.

v	t	$V = \log_e v$	$T = \log_e t$	VT	T^2
350	61	5.8579	4.1109	24.0812	16.8999
400	26	5.9915	3.2581	19.5209	10.6152
500	7	6.2146	1.9459	12.093	3.7865
600	2.6	6.3969	0.9555	6.1122	0.913
		$\sum V = 24.4609$	$\sum T = 10.2704$	$\sum VT = 61.8073$	$\sum T^2 = 32.2142$

The normal equations becomes

$$24.4609 = 4A + 10.2704b$$

$$61.8073 = 10.2704A + 32.2142b$$

On solving, we have $A = 6.5539$ & $b = -0.1709$

$$-A = \log_e a \Rightarrow a = e^A \Rightarrow a = e^{6.5539} = 701.9766$$

Then the required curve of fit is $y = (701.9766)e^{-0.1709x}$

⑨ Fit a second degree parabola to the following data.

x	1	1.5	2	2.5	3	3.5	4
y	1.1	1.3	1.6	2	2.7	3.4	4.1

→ Let $y = a + bx + cx^2$ be the second degree parabola

The associated equations are,

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \quad (n=7)$$

The relevant table is as follows.

x	y	xy	x^2y	x^2	x^3	x^4
1	1.1	1.1	1.1	1	1	1
1.5	1.3	1.95	2.925	2.25	3.375	5.0625
2	1.6	3.2	6.4	4	8	16
2.5	2	5	12.5	6.25	15.625	39.0625
3	2.7	8.1	24.3	9	27	81
3.5	3.4	11.9	41.65	12.25	42.875	150.0625
4	4.1	16.4	65.6	16	64	256
$\Sigma x = 17.5$	$\Sigma y = 16.2$	$\Sigma xy = 47.65$	$\Sigma x^2y = 154.475$	$\Sigma x^2 = 50.75$	$\Sigma x^3 = 161.875$	$\Sigma x^4 = 48.1875$

The normal equations become,

$$16.2 = 7a + 17.5b + 50.75c$$

$$47.65 = 17.5a + 50.75b + 161.875c$$

$$154.475 = 50.75a + 161.875b + 548.1875c$$

On solving, we obtain.

$$a = 1.0357 \approx 1.036$$

$$b = -0.1928 \approx -0.193$$

$$c = 0.2428 \approx 0.243$$

Thus the required second degree parabola of fit is

$$\boxed{y = 1.036 - 0.193x + 0.243x^2}$$

///

* CORRELATION

The process of establishing a relationship or connecting between two or more things, also if the change in one variable affects the change in other variables are said to be correlated.

* Types of correlation

① Positive Correlation : If an increase or decrease in the value of one variable corresponds to an increase or decrease in the other.

eg:- Height or weight, Rainfall & Yield of crops etc,

② Negative correlation : In an increase or decrease in one corresponds to decrease or increase on the other hand.

eg :- Increase in price of material & decrease in its demand.

* Coefficient of correlation :

Coefficient of correlation is denoted by r between two variables x & y and is defined by the relation.

Correlation the numerical measure of correlation between two variables x & y is known as the Pearson's coefficient of correlation.

* Correlation coefficient
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y} \quad (21) \quad \frac{\sum xy}{n \sigma_x \sigma_y}$$

where $\bar{x}$ = mean of x
 $\bar{y}$ = mean of y
 n = no. of values
 σ_x = Standard deviation of x
 σ_y = Standard deviation of y

$$x = x - \bar{x}$$

$$y = y - \bar{y}$$

* This can be put in an alternative form as follows.

If $X = x - \bar{x}$, $Y = y - \bar{y}$ we can write.

$$\sigma_x^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{\sum X^2}{n}$$

$$\sigma_y^2 = \frac{\sum (y - \bar{y})^2}{n} = \frac{\sum Y^2}{n}$$

$$\therefore \sigma_x \sigma_y = \frac{\sqrt{\sum X^2} \sqrt{\sum Y^2}}{n}$$

$$\therefore n \sigma_x \sigma_y = \sqrt{\sum X^2} \sqrt{\sum Y^2}$$

Then.

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$$

- * NOTE:-
- ① The co-efficient of correlation numerically does not exceed unity. $(-1 \leq r \leq 1)$
 - ② If $r = \pm 1$ we say that x & y are perfectly correlated & If $r = 0$ we say that x & y are non correlated.
 - ③ If $r = +1$, positive correlation.
 If $r = -1$, negative correlation.
 If $r = 0$, no correlation.

* Formula for the correlation coefficient r

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

proof :- let $z = x - y$

$$\therefore \frac{\sum z}{n} = \frac{\sum x}{n} - \frac{\sum y}{n}$$

$$\text{i.e. } \bar{z} = \bar{x} - \bar{y}$$

Hence, $(z - \bar{z}) = (x - y) - (\bar{x} - \bar{y})$

$$(z - \bar{z}) = (x - \bar{x}) - (y - \bar{y})$$

squaring both sides, taking summation and dividing by n ,

we have,
$$\frac{\sum (z - \bar{z})^2}{n} = \frac{\sum (x - \bar{x})^2}{n} + \frac{\sum (y - \bar{y})^2}{n} - \frac{2\sum (x - \bar{x})(y - \bar{y})}{n}$$

$$\text{i.e. } \sigma_z^2 = \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y,$$

since $r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$

$$\therefore \sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y.$$

Thus,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

* note :- In general if $z = ax + by$ we can obtain as before,

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2rab\sigma_x\sigma_y$$

$$\text{i.e. } \sigma_{ax+by}^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2rab\sigma_x\sigma_y.$$

* REGRESSION

Regression is an estimation of one independent variable from the other.

(or)

It is a method used for estimating the known values using the known values.

* Lines of Regression:

A line of regression is a straight line which gives the best fit in the least square to the given frequency. There are two types:

① Line of regression of x on $y \Rightarrow x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$

② Line of regression of y on $x \Rightarrow y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$

* Regression co-efficient:

$r \frac{\sigma_y}{\sigma_x}$ and $r \frac{\sigma_x}{\sigma_y}$ are known as regression co-efficients.

Their product will be equal to r^2 .

$$\text{i.e. } r \frac{\sigma_x}{\sigma_y} \cdot r \frac{\sigma_y}{\sigma_x} = r^2$$

$$r^2 = \text{coefficient of } x \cdot \text{coefficient of } y$$

$$r = \pm \sqrt{\text{coefficient of } x \cdot \text{coefficient of } y}$$

The sign of r will be +ve or -ve according to the regression coefficients are +ve or -ve.

* note:- ① the lines of regression are also of the form. (24)

$$Y = \frac{\sum XY}{\sum X^2} (X) \quad \& \quad X = \frac{\sum XY}{\sum Y^2} (Y)$$

where $X = x - \bar{x}$ & $Y = y - \bar{y}$.

this form will be useful to find out the coefficient of correlation by 1st obtaining the lines of regression as we have deduced that $r = \pm \sqrt{(\text{coeff of } x)(\text{coeff of } y)}$

② to compute the coefficient of correlation we prefer to use the formula. $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{xy}^2}{2\sigma_x\sigma_y}$

$$\therefore \sigma^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

* LIST OF FORMULAS

① correlation coefficient $r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n\sigma_x\sigma_y}$ (or) $\frac{\sum xy}{n\sigma_x\sigma_y}$

$$\sigma_x^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{\sum x^2}{n}, \quad \sigma_y^2 = \frac{\sum (y - \bar{y})^2}{n} = \frac{\sum y^2}{n}$$

③ regression line of x on $y \Rightarrow (x - \bar{x}) = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$

Regression line of y on $x \Rightarrow (y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (\bar{x} - \bar{y})$

④ $\bar{x} = \frac{\sum x}{n}, \quad \bar{y} = \frac{\sum y}{n}$

⑤ regression coefficient of x & y

$$r = \pm \sqrt{(\text{coefficient of } x)(\text{coefficient of } y)}$$

* PROBLEMS

- ① compute the coefficient of correlation and the equation of line of regression for the data,

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

→ we have, $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$ — (1)

we shall prepare the table showing the columns x, y, z , x^2, y^2, z^2 where $z = x - y$ so as to compute.

x	y	$z = x - y$	x^2	y^2	z^2
1	9	-8	1	81	64
2	8	-6	4	64	36
3	10	-7	9	100	49
4	12	-8	16	144	64
5	11	-6	25	121	36
6	13	-7	36	169	49
7	14	-7	49	196	49
$\Sigma x = 28$	$\Sigma y = 77$	$\Sigma z = -49$	$\Sigma x^2 = 140$	$\Sigma y^2 = 875$	$\Sigma z^2 = 347$

$$\bar{x} = \frac{\Sigma x}{n}, \quad \bar{y} = \frac{\Sigma y}{n}, \quad \bar{z} = \frac{\Sigma z}{n}, \quad n = 7$$

$$\therefore \bar{x} = \frac{28}{7} = 4, \quad \bar{y} = \frac{77}{7} = 11, \quad \bar{z} = \frac{-49}{7} = -7$$

$$\sigma_x^2 = \frac{\sum x^2}{n} - (\bar{x})^2, \quad \sigma_y^2 = \frac{\sum y^2}{n} - (\bar{y})^2$$

$$\sigma_x^2 = \frac{140}{7} - (4)^2 = 4, \quad \sigma_y^2 = \frac{875}{7} - (11)^2$$

$$\sigma_z^2 = \frac{\sum z^2}{n} - (\bar{z})^2 \quad \boxed{\sigma_y^2 = 4}$$

$$= \frac{347}{7} - (-7)^2$$

$$\boxed{\sigma_z^2 = 0.57}$$

we have, $\sigma_x^2 = 4$, $\sigma_y^2 = 4$, $\sigma_z^2 = \sigma_{x-y}^2 = 0.57$

substituting these values in ①

we have,

$$r = \frac{4+4-0.57}{2\sqrt{4}\sqrt{4}} = 0.92875 \approx 0.93$$

$$\text{Then } \boxed{r = 0.93}$$

* the lines of regression are given by,

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad \& \quad x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$y - 11 = \frac{(0.93)(2)}{2} (x - 4) \quad ; \quad x - 4 = \frac{(0.93)(2)}{2} (y - 11)$$

$$y - 11 = 0.93(x - 4) \quad ; \quad x - 4 = 0.93(y - 11)$$

$$\text{Then, } \boxed{y = 0.93x + 7.28} \quad \& \quad \boxed{x = 0.93y - 6.23} \text{ are}$$

the lines of regression.

2. Obtain the lines of regression and hence find the co-efficient of correlation for the data,

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

→ In this problem we first compute $\bar{x}, \bar{y}$ and denote $X = x - \bar{x}$, $Y = y - \bar{y}$ to form the table, inclusive of the columns XY, X^2, Y^2 .

Here $\bar{x} = 4$, $\bar{y} = 11$, $\therefore X = x - 4$, $Y = y - 11$

The relevant table is as follows.

x	y	X	Y	XY	X^2	Y^2
1	9	-3	-2	6	9	4
2	8	-2	-3	6	4	9
3	10	-1	-1	1	1	1
4	12	0	1	0	0	1
5	11	1	0	0	1	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
$\Sigma x = 28$	$\Sigma y = 77$			$\Sigma XY = 26$	$\Sigma X^2 = 28$	$\Sigma Y^2 = 28$

We shall consider regression lines in the form:

$$Y = \frac{\Sigma XY}{\Sigma X^2} \cdot X \quad \text{and} \quad X = \frac{\Sigma XY}{\Sigma Y^2} \cdot Y$$

$$\text{i.e. } y - 11 = \frac{26}{28} (x - 4) \quad ; \quad x - 4 = \frac{26}{28} (y - 11)$$

$$y - 11 = 0.93 (x - 4) \quad ; \quad x - 4 = 0.93 (y - 11)$$

Thus, $\boxed{y = 0.93x + 4.28}$ and $\boxed{x = 0.93y - 6.23}$

These are the regression lines and we compute r as the geometric mean of the regression co-efficients.

i.e., $r = \sqrt{(\text{coeff of } x)(\text{coeff of } y)} = \sqrt{(0.93)(0.93)} = 0.93$

The sign of ' r ' must be positive since both the regression coefficients are positive.

Thus, $\boxed{r = 0.93}$

Remark
(or)

We can independently compute r by using the formula

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}} = \frac{.26}{\sqrt{28} \sqrt{28}} = \frac{.26}{28} \approx 0.93$$

Thus, $\boxed{r \approx 0.93}$

Ex. 11.1 P

- (3) Find the correlation coefficient and the equation of the lines of regression for the following values of x & y .

x	1	2	3	4	5
y	2	5	3	8	4

→ The table showing the column of x, y, z, x^2, y^2 where $z = x - y$ is as follows.

Here $n = 5$.

x	y	$z = x - y$	x^2	y^2	z^2
1	2	-1	1	4	1
2	5	-3	4	25	9
3	3	0	9	9	0
4	8	-4	16	64	16
5	7	-2	25	49	4
$\Sigma x = 15$	$\Sigma y = 25$	$\Sigma z = -10$	$\Sigma x^2 = 55$	$\Sigma y^2 = 151$	$\Sigma z^2 = 30$

$$\bar{x} = \frac{\Sigma x}{n} = 3, \quad \bar{y} = \frac{\Sigma y}{n} = 5, \quad \bar{z} = \frac{\Sigma z}{n} = -2$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 = \frac{55}{5} - (3)^2 = 2$$

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2 = \frac{151}{5} - (5)^2 = 5.2$$

$$\sigma_z^2 = \frac{\Sigma z^2}{n} - (\bar{z})^2 = \frac{30}{5} - (-2)^2 = 2$$

we have, $\sigma_x^2 = 2$, $\sigma_y^2 = 5.2$, $\sigma_z^2 = \sigma_{x-y}^2 = 2$

$$\text{now, } r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

$$\text{Hence } r = \frac{2 + 5.2 - 2}{2\sqrt{2}\sqrt{5.2}} = 0.8062 \approx 0.81$$

Thus, $\boxed{r = 0.81}$ //

The equations of the regression lines are as follows.

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$; x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$y - 5 = (0.81) \frac{\sqrt{5.2}}{\sqrt{2}} (x - 3)$$

$$; x - 3 = \frac{(0.81)\sqrt{2}}{\sqrt{5.2}} (y - 5)$$

$$y - 5 = 1.306 (x - 3)$$

$$; x - 3 = 0.502 (y - 5)$$

Given, $\boxed{y = 1.306x + 1.082}$ and $\boxed{x = 0.502y + 0.49}$

these are the lines of regression.

- (4) Find the lines of regression and hence find the co-efficient of correlation for the data.

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	26	16	10	32	32

$$\rightarrow \bar{x} = \frac{\sum x}{10} = \frac{70}{10} = 7, \quad \bar{y} = \frac{\sum y}{10} = \frac{150}{10} = 15$$

we denote, $X = x - \bar{x}$ & $Y = y - \bar{y}$.

i.e. $X = x - 7$, $Y = y - 15$

The relevant table is as follows.

x	y	X	Y	x^2	y^2	XY
1	8	-6	-4	36	49	42
3	6	-4	-9	16	81	36
4	10	-3	-5	9	25	15
2	8	-5	-7	25	49	35
5	12	-2	-3	4	9	6
8	16	1	1	1	1	1
9	16	2	1	4	1	2
10	10	3	-5	9	25	-15
13	32	6	17	36	289	102
15	32	8	17	64	289	136
$\Sigma x = 70$	$\Sigma y = 150$			$\Sigma x^2 = 204$	$\Sigma y^2 = 818$	$\Sigma XY = 360$

we have lines of regression in the form,

$$Y = \frac{\Sigma XY}{\Sigma x^2} X \quad ; \quad X = \frac{\Sigma XY}{\Sigma y^2} \cdot Y$$

$$\text{i.e. } y - 15 = \frac{360}{204} (x - 7) \quad ; \quad x - 7 = \frac{360}{818} (y - 15)$$

$$y - 15 = 1.76 (x - 7) \quad ; \quad x - 7 = 0.44 (y - 15)$$

$$\text{Thus, } \boxed{y = 1.76x + 2.68} \quad \& \quad \boxed{x = 0.44y + 0.4}$$

these are the lines of regression.

$$r = \sqrt{(\text{coeff. of } x)(\text{coeff. of } y)} = \sqrt{(1.76)(0.44)} = +0.88$$

The sign of 'r' is +ve since both regression coefficients are positive. Thus $\boxed{r = 0.88} //$

5. The following data gives the age of husband (x) & the age of wife (y) in years. Form the two regression lines and calculate the age of husband corresponding to 16 years age of wife.

x	36	23	27	28	28	29	30	31	33	35
y	29	18	20	22	27	21	29	27	29	28

→ we prepare the following table, here $n=10$.

x	y	$z = x - y$	x^2	y^2	z^2
36	29	7	1296	841	49
26	18	5	529	324	25
27	20	7	729	400	49
28	22	6	784	484	36
28	27	1	784	729	1
29	21	8	841	441	64
30	29	1	900	841	1
31	27	4	961	729	16
33	29	4	1089	841	16
35	28	7	1225	784	49
$\Sigma x = 300$	$\Sigma y = 250$	$\Sigma z = 50$	$\Sigma x^2 = 9138$	$\Sigma y^2 = 6414$	$\Sigma z^2 = 306$

$$\bar{x} = \frac{300}{10} = 30, \quad \bar{y} = \frac{250}{10} = 25, \quad \bar{z} = \frac{50}{10} = 5$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 = \frac{9138}{10} - (30)^2 = 13.8, \quad \sigma_x = 3.71$$

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2 = \frac{6414}{10} - (25)^2 = 16.4, \quad \sigma_y = 4.05$$

$$\sigma_z^2 = \frac{\sum z^2}{n} - (\bar{z})^2 = \frac{306}{10} - (5)^2 = 5.6$$

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2 \sigma_x \sigma_y} = 0.82. \quad \text{on sub \& simplification}$$

The regression lines are,

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad ; \quad x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$y - 25 = (0.82) \frac{(4.05)}{(3.715)} (x - 30) \quad ; \quad x - 30 = (0.82) \frac{3.715}{4.05} (y - 25)$$

$$\boxed{y = 0.894x - 1.82} \quad \& \quad \boxed{x = 0.752y + 11.2}$$

These are the lines of regression.

When $y = 16$ $x = ?$

$$x = 0.752(16) + 11.2 = 23.232 = 23$$

The husband's age is 23 years corresponding to wife age of 16 years.

⑥ Find the co-efficient of correlation for the following data.

x	10	14	18	22	26	30
y	18	12	24	6	30	36

→ We have, $\bar{x} = \frac{\sum x}{n} = \frac{120}{6} = 20,$

$$\bar{y} = \frac{\sum y}{n} = \frac{126}{6} = 21$$

Let $X = x - \bar{x}$ and $Y = y - \bar{y}$

that is, $X = x - 20$, $Y = y - 21$ & we prepare the following data table.

x	y	X	Y	X^2	Y^2	XY
10	18	-10	-3	100	9	30
14	12	-6	-9	36	81	54
18	24	-2	3	4	9	-6
22	6	2	-15	4	225	-30
26	30	6	9	36	81	54
30	36	10	15	100	225	150
$\Sigma x = 120$	$\Sigma y = 126$			$\Sigma X^2 = 280$	$\Sigma Y^2 = 630$	$\Sigma XY = 252$

$$r = \frac{\Sigma XY}{\sqrt{\Sigma X^2} \sqrt{\Sigma Y^2}} = \frac{252}{\sqrt{280} \times \sqrt{630}} = 0.6$$

Then, $\boxed{r = 0.6}$

Note :- Since $\bar{x}$ and $\bar{y}$ are integers we preferred this formula to compute r .

7.18
④ compute $\bar{x}$, $\bar{y}$ and r from the following equations of the regression lines. $2x + 3y + 1 = 0$, $x + 6y - 4 = 0$

→ Since regression lines pass through $(\bar{x}, \bar{y})$ we must have,

$$2\bar{x} + 3\bar{y} + 1 = 0$$

$$\bar{x} + 6\bar{y} - 4 = 0$$

Solving above equations, we obtain

$$\boxed{\bar{x} = -2} \quad \& \quad \boxed{\bar{y} = 1}$$

now to find r we must correctly identify the regression line of y on x and x on y so as to write down the regression coefficients.

Suppose we write,

$$2x + 3y + 1 = 0 \quad \text{as} \quad 3y = -2x - 1 \quad (\text{or}) \quad y = -\frac{2}{3}x - \frac{1}{3}$$

$$\& \quad x + 6y - 4 = 0 \quad \text{as} \quad x = -6y + 4,$$

the regression co-efficients will be respectively $-2/3$ & -6 .

$$\therefore r = \sqrt{(-2/3) \times (-6)} = \sqrt{4} = \pm 2.$$

This cannot be accepted as r lies between -1 & $+1$.

So we must change the given equations into other possible form.

$$\text{i.e. } 2x + 3y + 1 = 0$$

$$2x = -3y - 1 \quad (\text{or})$$

$$x = -3/2 y - 1/2$$

$$\& \quad x + 6y - 4 = 0$$

$$6y = -x + 4 \quad (\text{or})$$

$$y = -1/6 x + 2/3$$

now the regression coefficients will be $-3/2$ & $-1/6$ so that

$$r = \sqrt{(-3/2) \times (-1/6)} = \sqrt{1/4} = \pm 1/2$$

The sign of r must be negative as both the regression coefficients are negative & hence $\boxed{r = -0.5}$, $\Rightarrow \boxed{\bar{x} = -2, \bar{y} = 1, r = -0.5}$

- ⑧ In a partially destroyed lab record, only the lines of regression of y on x and x on y are available as $4x - 5y + 33 = 0$ and $20x - 9y = 107$ respectively.

→ Firstly we have to express the given equations appropriately in the form $y = ax + b$ & $x = cy + d$.

Consider, $4x - 5y + 33 = 0$ (or) $5y = 4x + 33$ (or)
 $y = (4/5)x + (33/5)$.

Also consider $20x - 9y = 107$ (or) $x = (9/20)y + (107/20)$

From these, the regression co-efficients are respectively $4/5$ & $9/20$.

Also the coefficient of correlation ' r ' is the regression co-efficients.

i.e. $r = \sqrt{(4/5)(9/20)} = \sqrt{9/25} = \pm 3/5$

∴ The required value of r is $+3/5$, since both the regression co-efficients are positive.

Thus $\boxed{r = 3/5 = 0.6}$

- ⑨ $8x - 10y + 66 = 0$ & $40x - 18y = 214$ are the two regression lines. Find the mean of x 's, y 's and the correlation coefficient. Find σ_y if $\sigma_x = 3$.

→ W.K.T. that regression line passes through $\bar{x}$ & $\bar{y}$

$$8\bar{x} - 10\bar{y} = -66$$

$$40\bar{x} - 18\bar{y} = 214$$

On solving above Eqn we get

$$\boxed{\bar{x} = 13} \text{ \& \ } \boxed{\bar{y} = 14}$$

To find the regression co-efficients,
we have to rewrite the equation of regression lines.

$$10y = 8x + 66 \quad (01)$$

$$y = 0.8x + 6.6 \quad \text{--- (1)}$$

$$40x = 18y + 214 \quad (01)$$

$$x = 0.45y + 5.35 \quad \text{--- (2)}$$

from (1) & (2), $r \frac{\sigma_y}{\sigma_x} = 0.8$

$$r \frac{\sigma_x}{\sigma_y} = 0.45$$

correlation coefficient $r = \sqrt{(0.8) \times (0.45)} = \pm 0.6$

Thus $\boxed{r = \pm 0.6}$

Since both the regression coefficients are positive.

Also, $\sigma_x = 3$ by data &

we have $r \frac{\sigma_y}{\sigma_x} = 0.8$ which gives

$$0.6 \sigma_y = 2.4$$

Thus $\boxed{\sigma_y = 4}$ //

(10) The regression line of y on x & x on y are
 $y = ax + b$; $x = cy + d$. Show that the ratio of SD_x
 of y on x is $\sqrt{a/c}$ and $\bar{x} = \frac{bc+d}{1-ac}$, $\bar{y} = \frac{ad+b}{1-ac}$

→ i.e. regression co-efficients, $r \frac{\sigma_y}{\sigma_x}$ & $r \frac{\sigma_x}{\sigma_y}$ are
 the co-efficients of x and y in the equation of
 regression lines.

$\therefore$ we have $r \cdot \frac{\sigma_y}{\sigma_x} = a$ & $r \frac{\sigma_x}{\sigma_y} = c$

$\div$ one by the other we have,

$$r \frac{\sigma_y}{\sigma_x} \div r \frac{\sigma_x}{\sigma_y} = \frac{a}{c}$$

i.e. $\sigma_y^2 / \sigma_x^2 = a/c$

Thus $\boxed{\sigma_y / \sigma_x = \sqrt{a/c}}$

we also know that the regression lines passes through
 $(\bar{x}, \bar{y})$ & hence we must have,

$$\bar{y} = a\bar{x} + b \quad \text{--- ①}$$

$$\bar{x} = c\bar{y} + d \quad \text{--- ②}$$

using ① & ② we have, $\bar{x} = c(a\bar{x} + b) + d$

i.e. $\bar{x}(1-ac) = bc + d \quad \therefore \boxed{\bar{x} = \frac{bc+d}{1-ac}}$

Similarly using ② in ① we have, $\bar{y} = a(\bar{y} + d) + b$

i.e. $\bar{y}(1-ac) = ad + b \quad \therefore \boxed{\bar{y} = \frac{ad+b}{1-ac}}$

Thus we have proved all the required results.

(11) Given.

	x-series	y-series
mean	18	100
S.D	14	20

and

$r = 0.8$, write down the equation of the lines of regression and hence find the most probable value of y when $x = 40$.

→ By data $\bar{x} = 18$ & $\bar{y} = 100$;

$$\sigma_x = 14 \quad \& \quad \sigma_y = 20$$

We have the equation of the regression lines,

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad \& \quad x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

i.e., $y - 100 = \frac{(0.8)(20)}{14} (x - 18)$; $x - 18 = \frac{(0.8)(14)}{20} (y - 100)$

$$y - 100 = 1.14 (x - 18) \quad ; \quad x - 18 = 0.56 (y - 100)$$

Then $\boxed{y = 1.14x + 79.48}$ & $\boxed{x = 0.56y - 38}$

These are the lines of regression.

When $x = 40$, we obtain from the 1st equation

$$y = 1.14(40) + 79.48$$

$$\boxed{y = 159.28}$$

(12) A person while calculating the co-efficient of correlation between two variables x, y from a set of 25 observations obtained the following results.

$$\Sigma x = 125, \quad \Sigma y = 100, \quad \Sigma xy = 508, \quad \Sigma x^2 = 650, \quad \Sigma y^2 = 460$$

But it was later found that the pairs of values

x	8	6
y	12	8

were wrongly copied as $x : 6 \ 8$
 $y : 14 \ 6$

obtain the correct value for the correlation coefficient.

→ Computation for wrong values is as below.

x	y	xy	x^2	y^2
6	14	84	36	196
8	6	48	64	36
Σ 14	20	132	100	232

computation for correct values.

x	y	xy	x^2	y^2
8	12	96	64	144
6	8	48	36	64
Σ 14	20	144	100	208

It may be observed that the value so $\Sigma x, \Sigma y, \Sigma x^2$ are unchanged even after the correction. However we have.

$$\text{correct } \Sigma xy = 508 - 132 + 144 = 520$$

$$\text{correct } \Sigma y^2 = 460 - 232 + 208 = 436.$$

Hence the corrected values of the mean & S.D. of x & y are.

$$\bar{x} = \frac{\Sigma x}{n} = \frac{125}{25} = 5 \quad \& \quad \bar{y} = \frac{\Sigma y}{n} = \frac{100}{25} = 4$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 = \frac{650}{25} - (5)^2 = 1 \quad \therefore \sigma_x = 1$$

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2 = \frac{436}{25} - (4)^2 = 1.44 \quad \therefore \sigma_y = 1.2$$

we have $r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y}$

$$= \frac{\sum xy - \sum x \bar{y} - \sum \bar{x} y + \sum \bar{x} \bar{y}}{n \sigma_x \sigma_y}$$

$$= \frac{1}{\sigma_x \sigma_y} \left[\frac{\sum xy}{n} - \frac{\bar{y} \sum x}{n} - \frac{\bar{x} \sum y}{n} + \frac{n \bar{x} \bar{y}}{n} \right]$$

$$= \frac{1}{\sigma_x \sigma_y} \left[\frac{\sum xy}{n} - \bar{y} \bar{x} - \bar{x} \bar{y} + \bar{x} \bar{y} \right]$$

$$r = \frac{\frac{\sum xy}{n} - \bar{x} \bar{y}}{\sigma_x \sigma_y} = \frac{\frac{520}{25} - (5)(4)}{(1)(1.2)} = 0.6666$$

Hence the corrected value of $r \approx 0.67$ //

- (13) If x & y are random variables with SDs σ_x & σ_y , it was found that random variables $(x+y)$, $(x-y)$, $(2x+y)$ respectively have variance 15, 11, 29.

compute SDs of x , y & also the coefficient of correlation.

→ Given $\sigma_{x+y}^2 = 15$, $\sigma_{x-y}^2 = 11$, $\sigma_{2x+y}^2 = 29$ — (1)

we have the formula,

$$\sigma_{ax+by}^2 = a^2 \sigma_x^2 + b^2 \sigma_y^2 + 2rab \sigma_x \sigma_y$$

Taking (a, b) respectively equal to $(1, 1)$, $(1, -1)$ & $(2, 1)$

we have,

$$\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 + 2r \sigma_x \sigma_y$$

$$\sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 - 2r \sigma_x \sigma_y$$

$$\sigma_{2x+3y}^2 = 4\sigma_x^2 + 9\sigma_y^2 + 12r\sigma_x\sigma_y$$

Using ① in these equations we have,

$$\sigma_x^2 + \sigma_y^2 + 2r\sigma_x\sigma_y = 15 \quad \text{--- (2)}$$

$$\sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y = 11 \quad \text{--- (3)}$$

$$4\sigma_x^2 + \sigma_y^2 + 4r\sigma_x\sigma_y = 29 \quad \text{--- (4)}$$

$$\textcircled{2} + \textcircled{3} \text{ gives, } 2\sigma_x^2 + 2\sigma_y^2 = 26$$

$$\text{or (1) } \sigma_x^2 + \sigma_y^2 = 13 \quad \text{--- (5)}$$

$$2 \times (3) + (4) \text{ gives } 6\sigma_x^2 + 3\sigma_y^2 = 51$$

$$2\sigma_x^2 + \sigma_y^2 = 17 \quad \text{--- (6)}$$

Solving ⑤ & ⑥ we obtain,

$$\sigma_x^2 = 4 \text{ \& } \sigma_y^2 = 9$$

$$\text{then } \sigma_x = 2 \text{ \& } \sigma_y = 3$$

Using these values in (2) we have,

$$4 + 9 + 12r = 15$$

$$\text{Then } r = 1/6$$

$$\boxed{r = 0.17} \text{ \%.}$$

(14) show that if θ is the angle between the lines of regression, then

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right)$$

→ n.b.t if θ is acute,

the angle between the lines $y = m_1 x + c_1$ and

$y = m_2 x + c_2$ is given by,

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

we have the lines of regression,

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \rightarrow \textcircled{1} \quad x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

we write the equation as

$$y - \bar{y} = \frac{\sigma_y}{r \sigma_x} (x - \bar{x}) \rightarrow \textcircled{2}$$

slopes of $\textcircled{1}$ & $\textcircled{2}$ are respectively given by,

$$m_1 = \frac{r \sigma_y}{\sigma_x} \text{ and } m_2 = \frac{\sigma_y}{r \sigma_x}$$

substituting them in formula for $\tan \theta$ we have,

$$\tan \theta = \frac{\frac{\sigma_y}{r \sigma_x} - \frac{r \sigma_y}{\sigma_x}}{1 + \frac{r \sigma_y}{\sigma_x} \cdot \frac{\sigma_y}{r \sigma_x}}$$

$$\tan \theta = \frac{\frac{\sigma_y}{\sigma_x} \left(\frac{1}{r} - r \right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}} = \frac{\frac{\sigma_y}{\sigma_x} \left(\frac{1-r^2}{r} \right)}{\frac{\sigma_x^2 + \sigma_y^2}{\sigma_x^2}}$$

$$\boxed{\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left[\frac{1-r^2}{r} \right]}$$

* note :-

(1) If $r = \pm 1$, $\tan \theta = 0 \therefore \theta = 0$, which implies that the two regression lines coincide & hence the variables are perfectly correlated.

(2) Also if $r = 0$, $\tan \theta = \infty$ (or) $\theta = \pi/2$.

This implies that the lines are perpendicular & hence the variables are uncorrelated.

(15) If the co-efficient of correlation between two variables x and y is 0.5 & the acute angle between their lines of regression is $\tan^{-1}(3/5)$ show that $\sigma_y = 2\sigma_x$ (or) $\sigma_x = 2\sigma_y$.

→ By data $r = 0.5$ & $\theta = \tan^{-1}(3/5)$ or $\tan \theta = 3/5$

The angle b/w the lines of regression is given by,

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right)$$

$$\therefore \frac{3}{5} = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left\{ \frac{1 - (1/2)^2}{1/2} \right\}$$

$$\text{i.e., } \frac{3}{5} = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{3}{2} \right)$$

$$\frac{1}{5} = \frac{\sigma_x \sigma_y}{2(\sigma_x^2 + \sigma_y^2)}$$

$$\text{i.e., } 2\sigma_x^2 + 2\sigma_y^2 = 5\sigma_x \sigma_y$$

$$2\sigma_x^2 - 5\sigma_x \sigma_y + 2\sigma_y^2 = 0$$

Factorizing we obtain, $(2\sigma_x - \sigma_y)(\sigma_x - 2\sigma_y) = 0$

Thus we get $\boxed{\sigma_y = 2\sigma_x}$ (or) $\boxed{\sigma_x = 2\sigma_y}$ //

16) In a bivariate distribution $\sigma_x = \sigma_y$ & the angle between the regression lines is $\tan^{-1}(3)$. Find the correlation co-efficient.

→ If θ be the angle between the lines of regression we have

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right) \quad \text{--- (1)}$$

By data, $\theta = \tan^{-1}(3)$ & $\sigma_x = \sigma_y$

Hence (1) becomes,

$$3 = \frac{\sigma_x^2}{2\sigma_x^2} \left(\frac{1-r^2}{r} \right)$$

$$6r = 1-r^2$$

$$\text{i.e. } r^2 + 6r - 1 = 0$$

$$r = \frac{-6 \pm \sqrt{40}}{2}$$

$$= \frac{-6 \pm 2\sqrt{10}}{2}$$

$$= -3 \pm \sqrt{10}$$

$$\text{i.e. } r = +0.1623 \quad (\text{or}) \quad r = -6.1623$$

Since $|r| \leq 1$ the required $\boxed{r = 0.1623}$

* RANK CORRELATION and -An expression for the rank correlation coefficient.

The coefficient of correlation in respect of the ranks of some two characteristics of an individual or an observation is called the rank correlation coefficient usually denoted by ρ .

$$\rho = 1 - \frac{6 \sum (x-y)^2}{n(n^2-1)} \quad (or) \quad 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

* Note:-

① If the rankings of x, y are entirely in the same order like for example, $x = 1, 2, 3, 4, 5$,
 $y = 1, 2, 3, 4, 5$ then

$$\sum d^2 = \sum (x-y)^2 = 0$$

This will give us $\rho = +1$ & is called perfect direct correlation.

② If the ranking of x & y are entirely in the opposite order like for example, $x = 1, 2, 3, 4, 5$, $y = 5, 4, 3, 2, 1$

$$\text{then } \sum d^2 = \sum (x-y)^2 = (-4)^2 + (-2)^2 + (0)^2 + (2)^2 + (4)^2 = 40$$

This will give us, $\rho = 1 - \frac{6(40)}{5(5^2-1)} = -1$ & is called perfect inverse correlation.

③ ρ in case of repeated ranks.

→ If two or more magnitudes are repeated, then there will be a 'tie' for a particular rank. In such case we assign the average rank to all those magnitude & use the correction factor $\frac{m(m^2-1)}{12}$ along with $\sum d^2$ in the formula for ρ where m denotes the number of times a magnitude is repeated. The correction factor must be added every time the 'tie' occurs for a particular rank.

* PROBLEMS :

- ① Ten competitors in a beauty contest are ranked by two judges in the following order. Compute the co-efficient of rank correlation.

I	1	6	5	3	10	2	4	9	7	8
II	6	4	9	8	1	2	3	10	5	7

→ we have,
$$\rho = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

For the given data, $n=10$ & we have,

$$\begin{aligned} \sum d^2 &= (1-6)^2 + (6-4)^2 + (5-9)^2 + (3-8)^2 + (10-1)^2 \\ &\quad + (2-2)^2 + (4-3)^2 + (9-10)^2 + (7-5)^2 + (8-7)^2 \end{aligned}$$

i.e. $\sum d^2 = 25 + 4 + 16 + 25 + 81 + 0 + 1 + 1 + 4 + 1 = 158$

Hence,
$$\rho = 1 - \frac{6(158)}{10(10^2-1)} = 0.042$$

Thus,
$$\boxed{\rho = 0.042} //$$

- ② Ten competitors in music contest are ranked by 3 judges A, B, C in the following order. Use the rank correlation coefficient to decide which pair of judges have the nearest approach to common taste of music.

A	1	6	5	10	3	2	4	9	7	8
B	3	5	8	4	7	10	2	1	6	9
C	6	4	9	8	1	2	3	10	5	7

→ we shall compute ρ_{AB} , ρ_{BC} , ρ_{CA} with the help of the following table where d is the difference in ranks.

A	B	C	d_{AB}^2	d_{BC}^2	d_{CA}^2
1	3	6	4	9	25
6	5	4	1	1	4
5	8	9	9	1	16
10	4	8	36	16	4
3	4	1	16	36	4
2	10	2	64	64	0
4	2	3	4	1	1
9	1	10	64	81	1
7	6	5	1	1	4
8	9	7	1	4	1
			$\sum d_{AB}^2$ = 200	$\sum d_{BC}^2$ = 214	$\sum d_{CA}^2$ = 60

we have, $\rho = 1 - \frac{6 \sum d^2}{n(n^2-1)}$, $n=10$

$$\rho_{AB} = 1 - \frac{6(200)}{10(10^2-1)} = -0.21$$

$$\rho_{BC} = 1 - \frac{6(214)}{10(10^2-1)} = -0.294$$

$$\rho_{CA} = 1 - \frac{6(60)}{10(10^2-1)} = +0.636$$

It may be observed that ρ_{AB} & ρ_{BC} are negative which means their tastes are opposite.

Since ρ_{CA} is positive & is nearer to 1 (perfect correlation).

We conclude that the judges 'C' & 'A' have the nearest approach to common taste of music.

- ③ Ten students got the following percentage of marks in two subjects x & y . Compute their rank correlation coefficient.

Marks in x	78	36	98	25	75	82	90	62	65	39
Marks in y	84	51	91	60	68	62	86	58	53	47

→ We prepare the table consisting of the given data along with the ranks assigned according to their order of the magnitude.

In the subject x , 98 will be awarded rank 1.,

90 as rank 2 & so on - - -

Marks x	Rank(x)	Marks in y	Rank(y)	$d = (x - y)$	$d^2 = (x - y)^2$
78	4	84	3	1	1
36	9	51	9	0	0
98	1	91	1	0	0
25	10	60	6	4	16
75	5	68	4	1	1
82	3	62	5	-2	4
90	2	86	2	0	0
62	7	58	7	0	0
65	6	53	8	-2	4
39	8	47	10	-2	4
					$\Sigma d^2 = 30$

we have
$$r = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)} \quad , n = 10$$

$$r = 1 - \frac{6(30)}{10(10^2 - 1)}$$

$$r = 1 - \frac{6(30)}{10(10^2-1)}$$

$$r = 0.81818$$

Then $\boxed{r \approx 0.82}$ //

- (4) The coefficient of rank correlation obtained by ten students in Statistics & Accuracy was 0.2. It was later discovered that the difference in ranks in the two subjects of one of the students was wrongly taken as 9 instead of 7. Find the correct rank correlation coefficient.

→ we have, $r = 1 - \frac{6 \sum d^2}{n(n^2-1)}$

Here $n=10$ & $r=0.2$

$$\therefore 0.2 = 1 - \frac{6 \sum d^2}{10(10^2-1)}$$

$$\text{i.e. } \frac{6 \sum d^2}{990} = 1 - 0.2 = 0.8$$

$$\sum d^2 (\text{Incorrect}) = \frac{990 \times 0.8}{6} = 132$$

$$\text{Now correct } \sum d^2 = 132 - 9^2 + 7^2 = 100$$

$$\therefore \text{correct } r = 1 - \frac{6(100)}{10(10^2-1)} = 0.394$$

Then correct $\boxed{r = 0.394}$ //

⑤ Compute the rank correlation coefficient for the following data;

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	40

→ The data along with the assigned ranks is as below.

x	Rank x	y	Rank y	$d = x - y$	d^2
68	4	62	5	-1	1
64	6	58	4	-1	1
75	2.5	68	3.5	-1	1
50	9	45	10	-1	1
64	6	81	1	5	25
80	1	60	6	-5	25
75	2.5	68	3.5	-1	1
40	10	48	9	1	1
55	8	50	8	0	0
64	6	40	2	4	16
					$\Sigma d^2 = 72$

here in x the magnitude 75 is repeated twice for rank 2 & hence the average rank of 2 & 3 = 2.5 is assigned to both ($m=2$); Also in x the magnitude 64 is repeated thrice for rank 5 & hence the average rank of 5, 6, 7 = $\frac{5+6+7}{3} = 6$

is assigned to the magnitude 64 ($m=3$).

Further in y the magnitude 68 is repeated twice for rank 3 & hence the average rank of 3 & 4 = 3.5 is assigned to both of them ($m=2$).

(50)
We have the co-efficient of rank correlation for repeated rank.

$$r = 1 - \frac{6 \left[\sum d^2 + \frac{m(m^2-1)}{12} + \dots \right]}{n(n^2-1)}, \quad n=10$$

$$\text{if } r = 1 - \frac{6 \left[72 + \frac{2(2^2-1)}{12} + \frac{3(3^2-1)}{12} + \frac{2(2^2-1)}{12} \right]}{10(10^2-1)}$$

$$= 1 - \frac{6[72 + 0.5 + 2 + 0.5]}{990}$$

$$r = 0.545$$

$$\text{Thus } \boxed{r = 0.545} //$$

* STANDARD ERROR OF ESTIMATES

① The standard error of estimate of y on x is given by $S_y = \sigma_y \sqrt{1-r^2}$ &

Similarly standard error of estimate of x on y is given by $S_x = \sigma_x \sqrt{1-r^2}$

(or)

$$\textcircled{2} \quad S_{yx} = \sqrt{\frac{\sum (X - X_c)^2}{N}} \quad \text{where } X \rightarrow \text{Actual value}$$

$X_c \rightarrow \text{Estimated value.}$

$$S_{xy} = \sqrt{\frac{\sum (Y - Y_c)^2}{N}}$$

* PROBLEMS

① Find the standard error of estimates.

$$\sigma_x = 4 \quad \sigma_y = 2.2 \quad r = 0.8$$

→ Given $\sigma_x = 4$, $\sigma_y = 2.2$ & $r = 0.8$

Standard error of estimate of y on x is

$$S_{yx} = \sigma_y \sqrt{1-r^2}$$

$$S_{yx} = 2.2 \sqrt{1-(0.8)^2}$$

$$= 2.2 \sqrt{0.36}$$

$$= 2.2 (0.6)$$

$$\boxed{S_{yx} = 1.32}$$

x on y : is given by $S_{xy} = \sigma_x \sqrt{1-r^2}$

$$S_{xy} = 4 \sqrt{1-(0.8)^2}$$

$$= 4 \sqrt{0.36}$$

$$= 4 \times 0.6$$

$$\boxed{S_{xy} = 2.4}$$

⑨ From the following data calculate standard error of estimate.

X	6	2	10	4	8
X_c	4.7	2.1	9.9	6.0	7.3
Y	9	11	5	8	7
Y_c	8.0	10.6	5.4	9.3	6.7

→ We have S.E.E of y on x

$$S_{yx} = \sqrt{\frac{\sum (Y - Y_c)^2}{N}}$$

$$S_{xy} = \sqrt{\frac{\sum (X - X_c)^2}{N}}$$

where $N = 5$

X	X _c	(X-X _c)	(X-X _c) ²	Y	Y _c	(Y-Y _c)	(Y-Y _c) ²
6	4.9	1.3	1.69	9	8.0	1	1.00
2	2.1	0.1	0.01	11	10.6	0.4	0.16
10	9.9	0.1	0.01	5	5.4	0.4	0.16
4	6.0	0.2	0.04	8	9.3	1.3	1.69
8	7.3	0.7	0.49	7	6.7	0.3	0.09
ΣX=30	ΣX _c =30	Σ(X-X _c)=0	Σ(X-X _c) ² =6.20	ΣY=40	ΣY _c =40	Σ(Y-Y _c)=0	Σ(Y-Y _c) ² =3.1

now we have

x on y

$$S_{xy} = \sqrt{\frac{\sum (X-X_c)^2}{N}}$$

$$= \sqrt{\frac{6.20}{5}}$$

$$S_{xy} = \sqrt{1.24}$$

$$\boxed{S_{xy} = 1.11} //$$

y on x

$$S_{yx} = \sqrt{\frac{\sum (Y-Y_c)^2}{N}}$$

$$= \sqrt{\frac{3.1}{5}}$$

$$= \sqrt{0.62}$$

$$\boxed{S_{yx} = 0.787} //$$

3) Compute the two regression equations & calculate the standard error of estimates.

X	2	4	5	6	8	11
Y	18	12	10	8	7	5

→

$$\bar{x} = \frac{\sum X}{n} = \frac{36}{6} = 6$$

$$\bar{y} = \frac{\sum Y}{n} = \frac{60}{6} = 10$$

x	y	$X = x - 6$	$Y = y - 10$	X^2	Y^2	XY
2	18	-4	8	16	64	-32
4	12	-2	2	4	4	-4
5	10	-1	0	1	0	0
6	8	0	-2	0	4	0
8	7	2	-3	4	9	-6
11	5	5	-5	25	25	-25
$\Sigma x = 25$	$\Sigma y = 60$			$\Sigma X^2 = 50$	$\Sigma Y^2 = 106$	$\Sigma XY = -67$

Equation of Regression lines are

$$Y = \frac{\Sigma XY}{\Sigma X^2} X$$

$$X = \frac{\Sigma XY}{\Sigma Y^2} Y$$

$$y - \bar{y} = \frac{\Sigma XY}{\Sigma X^2} (x - \bar{x})$$

$$x - \bar{x} = \frac{\Sigma XY}{\Sigma Y^2} (y - \bar{y})$$

$$y - 10 = \frac{-67}{50} (x - 6)$$

$$x - 6 = \frac{-67}{106} (y - 10)$$

$$y - 10 = -1.34(x - 6)$$

$$x - 6 = -0.632(y - 10)$$

$$y - 10 = -1.34x + 8.04$$

$$x - 6 = -0.632y + 6.32$$

$$\boxed{y = -1.34x + 18.04}$$

$$\boxed{x = -0.632y + 12.32}$$

now to find standard error of estimate.

we have to construct a table.

X	Y	x_c	$(x - x_c)^2$	y_c	$(y - y_c)^2$
2	18	0.944	1.195	15.36	6.97
4	12	4.736	0.541	12.68	0.46
5	10	6.0	1	11.34	1.795
6	8	7.264	1.597	10	4
8	7	7.896	0	7.32	0.102
11	5	9.16	3.38	3.3	2.89
			7.633	$\sum (y - y_c)^2 = 16.217$	

To find x_c values,

consider eqn of regression line

$$(1) \quad x = -0.632y + 12.32$$

$$y = -1.34x + 18.04$$

$$x = -0.632(18) + 12.32$$

$$y = -1.34(2) + 18.04$$

$$x_c = 0.944$$

$$y_c = 15.36$$

$$(2) \quad x = -0.632(12) + 12.32$$

$$y = -1.34(4) + 18.04$$

$$x_c = 4.736$$

$$y_c = 12.68$$

$$(3) \quad x = -0.632(10) + 12.32$$

$$y = -1.34(5) + 18.04$$

$$x_c = 6.0$$

$$y_c = 11.34$$

$$(4) \quad x = -0.632(8) + 12.32$$

$$y = -1.34(6) + 18.04$$

$$x_c = 7.264$$

$$y_c = 10$$

$$(5) \quad x = -0.632(7) + 12.32$$

$$y = -1.34(8) + 18.04$$

$$x_c = 7.896$$

$$y_c = 7.32$$

$$(6) \quad x = -0.632(5) + 12.32$$

$$y = -1.34(11) + 18.04$$

$$x_c = 9.16$$

$$y_c = 3.3$$

$\therefore$ standard error estimate

$$S_{xy} = \sqrt{\frac{\sum (x - x_c)^2}{N}} = \sqrt{\frac{7.633}{6}}$$

$$S_{xy} = \sqrt{1.272}$$

(5)

$$S_{xy} = 1.127$$

$$S_{yx} = \sqrt{\frac{\sum (y - y_c)^2}{n}}$$

$$= \sqrt{\frac{16.217}{6}}$$

$$= \sqrt{2.702}$$

$$S_{yx} = 1.643$$

//

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